Chapter 19

Conventional bridges

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CONVENTIONAL BRIDGES OR IMPLANTS?

The advent of implant supported restorations has, to some extent, reduced the need for conventionally prepared bridges. However, the financial cost of implant retained restorations is higher than conventional bridges, due to the additional cost of the implant components. Others claim that there is little long-term difference due to the biological impact conventional bridge preparation has to abutment teeth and the potential need for further treatment. Despite the cost, implants require surgical procedures demanding sufficient quantity and quality of bone (see Chapter 17). These issues, together with the status of the remaining dentition, may influence the choice as to how to replace teeth and, as such, conventional bridges remain useful options in prosthodontics.

TERMINOLOGY

The terminology used for bridges or fixed partial prostheses/dentures includes the following (Figure 19.1):

- *Abutment*. The tooth or teeth that support the bridge.
- *Retainer*. The restoration that attaches the bridge to the abutment tooth or teeth.
- *Pontic*. The false tooth.
- *Connector*. The part of the bridge that connects the pontic to the retainer and is frequently overlooked.
- *Span*. The mesio-distal length of the edentulous space.

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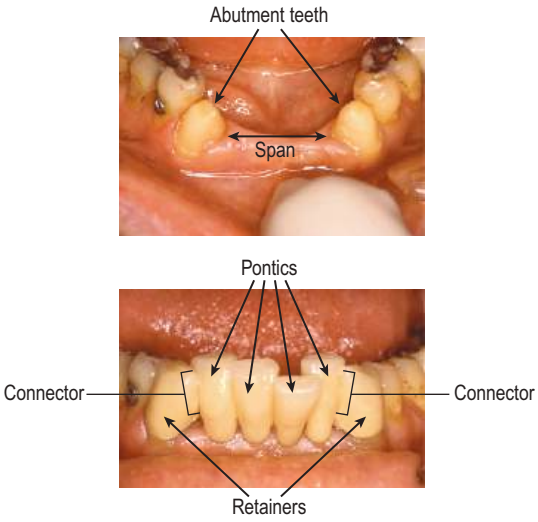


Figure 19.1 Terminology used in relation to bridges.



Figure 19.2 Patient with congenitally missing lateral incisor teeth and discoloured upper left central. Preparation of this tooth for a full coverage bridge retainer will also mask the discolouration.



Figure 19.3 The upper right central incisor space is larger than the upper left central incisor crown. Removal of crowns on the adjacent teeth will allow construction of a fixed–fixed bridge distributing the tooth sizes evenly and proportionally.

CONVENTIONAL VERSUS RESIN RETAINED BRIDGES

Conventional bridges are indicated when the abutment teeth are extensively restored and preparation of the tooth for an indirect restoration (usually a full coverage retainer or crown) is justified. As with a full coverage crown, the retainer has the additional benefit of protecting the remaining tooth tissue and core. Extensively restored teeth are usually unsuitable for minimal preparation (resin retained) bridges because of the presence and extent of the material–tooth interfaces.

Occasionally the shape and/or appearance of the abutment teeth may require alteration and this can be incorporated into the bridge design. The patient seen in Figure 19.2 has congenitally missing upper lateral incisor teeth. The image shows that the left central incisor is discoloured and the composite veneer used to mask this has not been successful. Preparation of the central incisor for a metal–ceramic retainer on a cantilever designed bridge will improve the appearance of the tooth and provide support for the pontic.

Preparation of teeth for full coverage crowns also allows some control over distribution of space and size of teeth. For example, Figure 19.3 shows the upper right central incisor tooth is missing, but the span (mesio-distal space) is greater than the mesio-distal width of the contralateral incisor crown. Removal of the crowns on the adjacent teeth will allow the retainers to be made with a slightly larger mesio-distal width and hide the space and tooth size discrepancy. In a similar way, tilted or angled abutment teeth can be aligned to some degree to improve the appearance in the anterior region.

Bridges with more than one pontic require a fixed–fixed or fixed–movable design rather than a cantilever design. This allows support and retention to be gained from the abutment teeth either side of the pontic. When the abutment teeth are minimally restored, minimal preparation (resin retained) bridges can be considered, but longer span bridges demand greater retention from the abutment teeth and this can only be achieved from conventional preparations. The risk of decementation of minimally prepared (resin retained) bridges becomes too great once the span increases by more than one pontic. For example, the edentulous span for the elderly patient seen in Figure 19.1 is too long for a minimally prepared (resin retained) bridge and the abutment teeth have been prepared for metal–ceramic retainers to optimize retention. The age of

the patient, the mild wear on the teeth and the exposed root surface will ensure sufficient secondary and tertiary dentine formation to reduce potential pulpal trauma during tooth preparation.

OPTIMIZING AESTHETICS

All bridges made in the smile line need to ensure an aesthetic symmetrical result. The space needed for the pontic has to mirror the ideal size of the tooth which it is to replace. The contralateral tooth should give a guide to the dimensions. If the space is too small, a number of solutions can be considered:

- The teeth either side can be increased in size by the addition of composites, veneers or crowns, depending on the restorative status of the teeth.
- The space can be closed orthodontically.
- The space can be increased orthodontically for the ideal sized tooth; however, there has to be a space for the adjacent tooth/teeth to be moved into and this often requires extraction of a tooth.
- Bridge designs showing some degree of tooth crowding can be used (Figures 19.1 and 19.4).



Figure 19.4 This patient has a crown on the upper right canine, and two separate cantilever conventional bridges with metal–ceramic retainers on the central incisors and lateral incisor pontics. There is a third minimum preparation (resin retained) cantilever bridge replacing the upper left canine using the first premolar tooth as an abutment.

When the space is too large compared to the contralateral tooth (Figure 19.3):

- The adjacent tooth/teeth can be crowned or recrowned to redistribute the size of the teeth and avoid any spacing or, if sound, composite can be added to the teeth.
- A cantilever bridge design can be considered, leaving residual spacing to one side of the pontic only.
- Orthodontic space closure to ideal size
- If the prosthetic tooth is to be placed in the centre of the space and residual spacing accepted, two options are available: a removable prosthesis or an implant.

BRIDGE DESIGN

The principles involved in the design of conventional and minimal preparation (resin retained) bridges are similar. The generic designs are cantilever, fixed–fixed and fixed–movable; the terminology is based on how the pontic is attached to the retainer(s).

Cantilever bridges

Cantilever bridges are designed so that the pontic is attached to a retainer on one side only, and are therefore usually more conservative of tooth tissue, especially if the other potential abutment tooth is minimally restored or sound. Retrievability is also easier if cantilever bridge designs are used. Consider the patient in Figure 19.4: three separate cantilever bridges have been placed instead of a fixed–fixed bridge design (see next section). Thus, if one abutment tooth fails due to secondary caries for example, only one tooth and bridge need replacing, so reducing the complexity of treatment and cost to the patient. Longer span bridges, particularly those where the pontic or retainers are involved in lateral guidance, increase the occlusal forces on the bridge and potentially reduce their survival. Historically, attempts were made to overcome this problem by using two abutment teeth. A typical example was a bridge to replace an upper canine; two premolar teeth were prepared for three-quarter crown retainers and linked to form a three unit cantilever bridge with a mesial pontic to replace the canine. This should be avoided as maintaining good oral hygiene beneath linked retainers is difficult, increasing the risk of caries and periodontal disease.

Occasionally cantilevered pontics can be used in conjunction with more extensive bridge designs. The patient seen in Figure 19.5 has a four unit fixed–fixed conventional bridge with the upper second premolar and second molar teeth used as abutments and the upper first premolar and molar teeth as pontics. The second premolar abutment has a pontic on either side and is referred to as a



Figure 19.5 Patient with a four unit fixed–fixed bridge design with the upper second premolar tooth as pier abutment and the first premolar cantilevered pontic.

pier abutment, the first premolar pontic being cantilevered off the bridge.

Spring cantilever bridges

Spring cantilever bridges are mainly of historic interest and were used when an anterior tooth was missing and the adjacent teeth were unrestored or where there was anterior spacing. Usually, molar teeth, which were extensively restored, were crowned and used to support a long palatal connector for the pontic in the anterior region. The introduction of dental implants has made these bridge designs obsolete; on rare occasions, however, clinicians may still see these bridges in service.

Fixed–fixed bridges

A fixed–fixed bridge has a pontic rigidly connected to retainers on both sides and has one path of insertion. The vertical orientation of both abutment teeth needs to be reasonably well aligned and parallel to each other (Figures 19.1, 19.5 and 19.6) to avoid undercuts. Most fixed–fixed bridges have full coverage crown retainers: if one abutment tooth had a relatively small restoration and an inlay was use as a retainer, occlusal contact on the tooth would lead to shear stresses being generated in the cement lute, with eventual debonding and risk of secondary caries (Figure 19.7). In theory, the use of a movable joint within the inlay would allow independent micromovement of that abutment tooth in an axial direction in line with the moveable joint (Figure 19.8).

The most important principle in planning tooth preparations on abutment teeth is retention. Preparations



Figure 19.6 Fixed–fixed anterior bridge preparation and cemented bridge. Note the abutment teeth are well aligned.

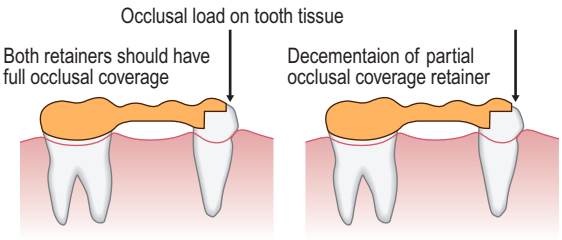


Figure 19.7 Diagram depicting an incorrect fixed–fixed bridge design linking a full occlusal coverage retainer on the molar tooth with a partial occlusal coverage retainer (inlay) on the premolar tooth. Occlusal loading on the premolar tooth generates shear stresses in the cement lute with eventual debonding of the inlay retainer. The retainers for fixed–fixed bridge designs should always have full occlusal coverage.

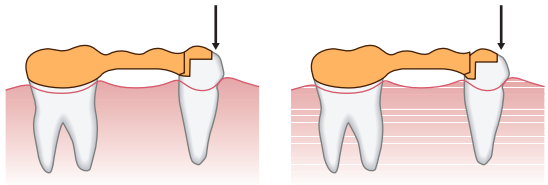


Figure 19.8 Incorporation of a moveable joint in the inlay retainer will allow some differential axial movement between the abutment teeth, providing the joint is in line with the long axis of the premolar tooth.

should follow the general principle of long and near parallel (at least 10–15° taper) sides. Fixed–fixed conventionally designed bridges present challenges to ensure that undercuts are not introduced between abutment teeth, and whilst anterior preparations are often easier because of direct vision, posterior ones are more challenging. Take the patient in Figure 19.9 for example: the first premolar tooth has been prepared with a labial shoulder and palatal chamfer to accept a full coverage metal–ceramic retainer. The first molar tooth is not in the smile line and a preparation for a full gold crown with a buccal and palatal chamfer finish to preserve as much of the tooth tissue as possible has been carried out. Note that no undercuts are evident between the two abutment preparations.

Fixed–movable bridges

A fixed–movable bridge has a pontic which is rigidly connected to a retainer at one end and to a moveable joint at the other. The retainer to which the pontic is rigidly attached is called the major retainer and the one that houses the movable joint is called the minor retainer (Figure 19.10). The size of the retainer has no bearing on

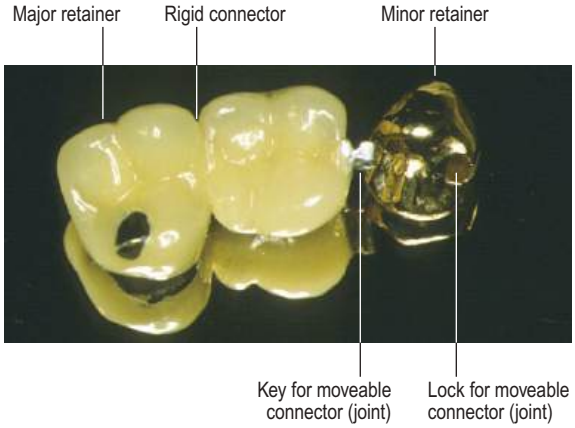


Figure 19.10 Components of a conventional fixed–movable designed bridge.



Figure 19.9 Fixed–fixed bridge with full coverage metal retainer on the upper left molar tooth and metal–ceramic retainer on the premolar tooth. Note metal occlusal coverage on both abutment teeth preserving the clinical crown height and improving retention.



Figure 19.11 The lower right second premolar tooth has been prepared for a three-quarter gold minor retainer. The lock can be seen in the distal aspect of the minor retainer. The key on the mesial aspect of the pontic seats into the lock of the minor retainer. The base of the lock resists mesial tilting of the molar abutment which has a full metal coverage major retainer.

the terminology. Normally, the movable joint is attached to the mesial retainer and is the smaller abutment tooth (Figure 19.11). The joint is usually placed in the mesial retainer as teeth tend to drift and tilt in a mesial direction. The potential for this to occur in a distal abutment tooth is resisted by the key pressing against the base of the movable joint (Figure 19.11 bottom two images). If care is taken with the occlusion and there is good interdigitation of teeth, tooth movement is unlikely to occur and the movable joint could be placed in the distal (larger) retainer.

The movable joint must have a base to achieve support for the pontic and must be within the retainer. Consider the bridge design for the patient in Figure 19.12. This design is wrong as the key is cantilevered off the retainer and whilst it sits in the lock in the mesial of the pontic, no support is given to the pontic. As a result, the molar pontic is essentially cantilevered off the distal abutment tooth and mesial tilt of this tooth has occurred with the key emerging from the lock.

Generally, fixed–movable bridges are indicated for posterior bridges where the space needed for the movable joint can be incorporated into the retainer without interfering with the appearance of the bridge. The major advantage of the fixed–movable bridge design is that the movable joint can overcome differences in angulation of abutment teeth; retainers on each abutment tooth can have optimum retention and path of insertion irrespective of the alignment of the abutment teeth (Figures 19.13 and 19.14).

The movable joint of a fixed–movable bridge is usually achieved by using factory-made plastic precision fitting components (Figure 19.15) which are incorporated into the laboratory wax-up of the bridge. Figure 19.16 shows the laboratory construction of a hybrid fixed–movable bridge. The minor retainer is a conventional full coverage metal–ceramic retainer, whilst the major retainer is a



Figure 19.12 Incorrect fixed–movable bridge design. The lock is incorrectly placed in the pontic and even if it has a base to it, it has not resisted mesial tilting of the major retainer and the key appears to be coming out of the lock.

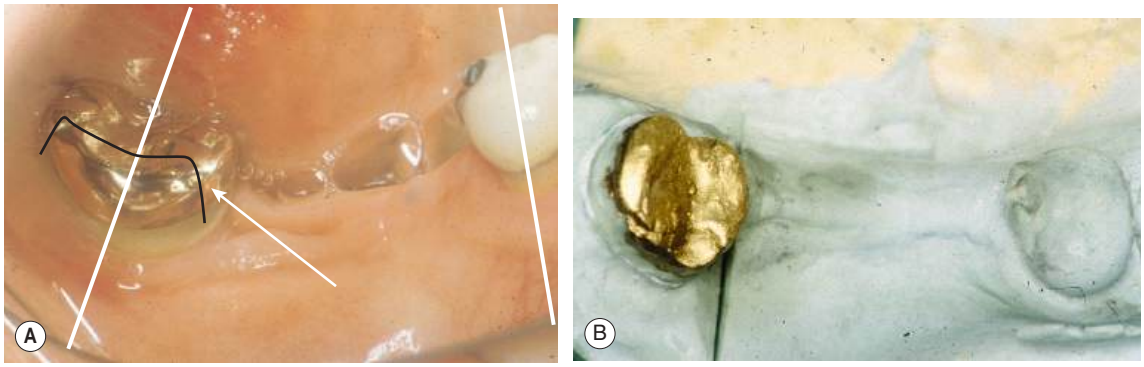


Figure 19.13 **A** Two potential abutment teeth which are poorly aligned. Any attempt at preparing the distal abutment tooth in alignment with the long axis of the mesial tooth for a fixed–fixed bridge design will lead to overpreparation of the mesial aspect of the tooth (arrowed). This may cause greater pulpal damage and an overtapered preparation, and hence poorer retention. **B** The working model of the patient seen in **A**. Note that the minor retainer has already been cemented and the lock is angled in line with the long axis of the distal abutment. The preparation on this tooth can therefore have the optimum taper and a different path of insertion to the minor retainer. Usually the entire bridge is made and cemented in the same appointment, obviously seating the minor retainer first and then the major retainer and pontic.



Figure 19.14 Another example of a fixed–movable bridge design allowing both abutment teeth to have optimal retention without the need for ensuring that they are parallel.

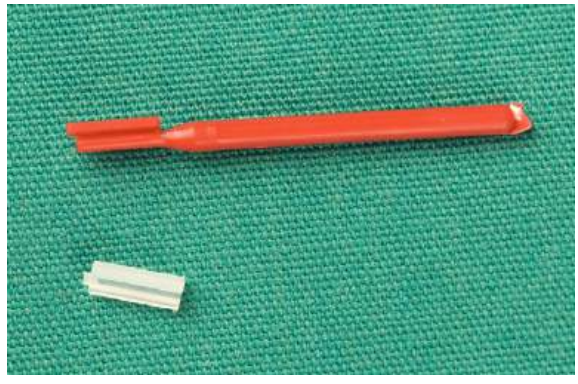


Figure 19.15 Factory-made Tube Lock joint (Sterngold Attachments, MA, USA).



Figure 19.16 Laboratory stages of making a fixed–movable hybrid bridge incorporating the Tube Lock joint (top six images) and when cemented intraorally.

minimal preparation (resin retained) inlay and wing (see Chapter 18).

There is a theoretical advantage that the presence of a movable joint between the retainers may allow some movement (called stress breaking) and reduce the effect of lateral forces applied to the abutments. However, since most joints are precision made (Figure 19.15), any movement between the major and minor connectors is unlikely to have any significant clinical effect and can only take place along the long axis of the joint.

PREPARATION FOR CONVENTIONAL BRIDGES

Most preparations for conventional bridges will involve full coverage retainers. When preparing teeth for fixed–fixed bridge designs, it is often better to prepare the teeth together to avoid introducing undercuts between abutments. For example, when preparing the axial walls it is advisable to visualize the long axis of each abutment

tooth and check that the preparations are aligned without overpreparing either tooth. Preparing corresponding surfaces a little at a time and frequently moving from one preparation to another without altering the angle of the tapered bur ensures that the preparations are better aligned with no undercuts; the longer the span, the greater the difficulty in keeping preparations aligned. It is equally important not to alter the angle of the dental mirror when moving it to visualize one preparation and then another.

PONTIC DESIGN

The design concepts for pontics are the same for minimum preparation (resin retained) bridges and conventional bridges and should always be prescribed to the technician. The different pontic designs are:

- Wash through or sanitary (Figure 19.17)
- Saddle (Figure 19.17)
- Dome (Figure 19.17)
- Ridge lap (Figure 19.17)
- Modified ridge lap (Figure 19.17)
- Ovate.

When a bridge is removed, the underlying edentulous ridge often appears red and inflamed, and this has been attributed to poor oral hygiene and the presence of a bacterial biofilm in contact with the surface of the alveolar ridge. When bridge hygiene is poor the inflamed tissue can often become hyperplastic in nature (Figures 19.18 and 19.19). The regular use of super floss beneath the pontic is necessary and should remove the plaque, preventing this from occurring. Some argue that the ridge appears red because it has become less keratinized as the tissue is protected from the frictional contacts made during mastication. In most situations it would be preferable to use a more cleansable pontic design.

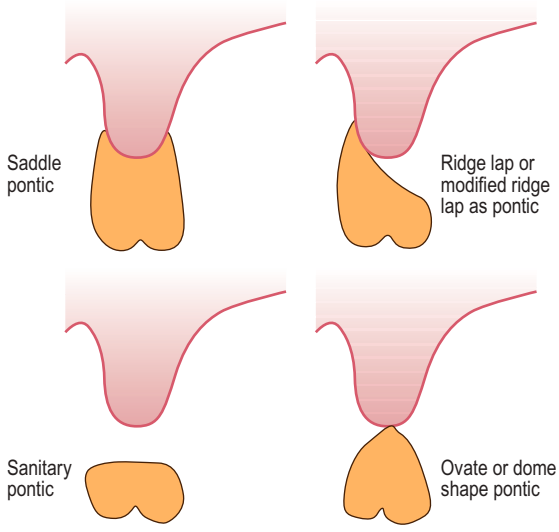


Figure 19.17 Diagrammatic representation of a section through pontics of different designs and their relationship to the edentulous ridge.



Figure 19.18 This patient's bridge has decemented. Oral hygiene is poor generally, but where the pontic sat the tissues are inflamed and becoming hyperplastic.



Figure 19.19 A saddle pontic in situ. Super flossing beneath this pontic has been difficult and the gingival tissues are inflamed and swollen.

The wash through or sanitary pontic (Figure 19.20) is only appropriate to replace posterior teeth which are not seen in the smile line. This historical bridge design ensures that there is no contact beneath the pontic and the edentulous ridge, and that there is sufficient space beneath the pontic not to trap food, which is easily 'washed away'. Whilst not often used, this design is useful when needed to provide a functional and stable occlusion and where oral hygiene procedures need to be kept simple. Adequate oral hygiene can be achieved with the use of an interdental bottle brush, such as a TePe brush.



Figure 19.20 Wash through or sanitary pontic used to achieve a stable occlusion and improve function.



Figure 19.21 Saddle pontic.

The most popular choice from a patient's perspective is the saddle pontic which has a similar profile to natural teeth (Figure 19.21). The inevitable scalloped shape of the surface along the edentulous ridge buccally and lingually provides a more realistic 'feel' to the bridge but its concaved ridge surface makes it more difficult to clean than the ridge lap designs.

The ridge lap and modified ridge lap designs, like the saddle-shaped pontic, provide a conventional buccal contour to create a tooth pontic that has a realistic appearance visible on the buccal surface but is cut back on the palatal or lingual surface to allow better access for cleaning (Figures 19.22 and 19.23). The modified ridge lap has an even more extensive removal of the palatal surface



Figure 19.22 Ridge lap pontic on a conventional fixed–fixed designed bridge.



Figure 19.23 Palatal view of a modified ridge lap pontic having minimal contact with the edentulous ridge buccally.

to allow greater access for cleaning with super floss and other oral hygiene products.

The dome-shaped pontic is used when aesthetics at the gingival margin is not important. It has a convex ridge surface which makes point contact with the crest of the edentulous ridge, again keeping contact with the ridge to a minimum and facilitating oral hygiene procedures.

The ovate pontic is an uncommon design which is used in implant restored dentitions where extensive edentulous ridges make the creation of an interdental papilla between adjacent pontics or abutment teeth difficult. The ovate pontic is tooth shaped with the addition of a bulbous end, shaped like a short root, which forms by pressure an indentation into the edentulous ridge. The shaping of the ridge is normally produced during the provisional bridge construction and when the definitive bridge has been produced the indentation gives the illusion of an interdental papilla. Electrosurgery has also been suggested to remove soft tissue from the edentulous ridge, recontouring it to receive an ovate pontic. This gives an emergence profile to the pontic comparable to that of a natural tooth (Elder & Djemal, 2008).

THE CONNECTOR

Little attention is often paid to the connector between the pontic and the retainer but this is an important aspect to the bridge. The connector has to be of an adequate dimension for strength and rigidity, but it is important that it is not too large so as to encroach upon the gingival margin of the abutment tooth. This can be a common mistake because the laboratory technician sections the model to remove the individual dies. In order to contour the cervical aspect of the retainer correctly and to produce an acceptable emergence profile without ledges, the technician trims away the gingival area from the die (see Figure 19.5). Reference to the gingival area is therefore lost and the connector can become too large (Figure 19.24). At try-in it is important to check this aspect of the bridge and ensure that oral hygiene procedures are possible.



Figure 19.24 The connector between the pontic and retainer for the cantilever bridge on the upper right canine is too large, making oral hygiene difficult and resulting in periodontal problems (arrowed).

THE OCCLUSION

The aim of a bridge is to replace teeth and provide function. However, loading abutment teeth with pontics will increase the forces transmitted to the periodontium. Therefore, occlusal contacts on bridges need to be carefully controlled. The abutment teeth and pontics should have contacts in the intercuspal position but excursive contacts should only be directed onto the abutment teeth when necessary. If guidance is also established on the pontics, damaging lateral forces may occur, especially with cantilever bridge designs, increasing the risk of failure.

In a Class I incisor relationship canine guidance causes posterior disclusion. This immediate disclusion of the posterior teeth simplifies the construction of posterior bridges as it removes any need to ensure that the palatal cusps on the upper teeth and the buccal cusps on the lower teeth meet during lateral excursions. The immediate disclusion of the teeth in canine guidance in a Class I incisor relationship means that the cusps can be made to their optimum height. During group function the need to have multiple tooth contacts during lateral excursion complicates the construction of the occlusal shape of bridges and increases the load transferred to the abutment teeth.

How many teeth can we replace with a bridge?

This is a frequently asked question and historically dentists referred to 'Ante's Law' (1926) which states that 'the total periodontal membrane area of the abutment teeth must equal or exceed that of the teeth to be replaced' (Lulic et al., 2007). This law gives an oversimplified guide to the clinician; however, it is based on engineering and not biological principles and is far too dogmatic. Teeth differ in root length, coronal size and restorative status, and occlusal forces differ between individuals and occlusal schemes. All of these factors have a potential impact on the number of teeth that can be replaced and should be taken into consideration when planning a bridge. By virtue of periodontal mechanoreceptors, proprioceptors and pain receptors, patients are able to adapt to changes in the occlusal load placed on abutment teeth, and in certain circumstances more extensive bridges can be made, providing care is taken with the occlusion (Figure 19.25).

LONGEVITY OF CONVENTIONAL BRIDGES

The data to support the lifespan of conventional bridges are not extensive but suggest that the survival is longer than minimal preparation (resin retained) bridges.



Figure 19.25 Longer edentulous spans where good periodontal bone support means that more extensive bridges could be considered, taking care with the occlusion.

A number of systematic reviews indicate that the long-term expectation of conventional fixed–fixed bridges is over 10 years and at around 15 years the increase in annual failure rate suggests that they become less viable (Pjetursson et al., 2004). These figures compare well with current data for implant supported restorations. The data from these studies also suggest that bridges fail within two main categories: decementation in the early stages and later mechanical failure of the retainer or pontic. Failure due to caries or periodontal disease is less common. The other major finding from systematic reviews is that fixed–fixed designs have a better prognosis than cantilevers. Fixed–movable designs are not as commonly used outside the UK as other forms and so comparison is not possible. For comparison, the longevity of minimal preparation (resin retained) bridges is around 5–7 years.

EXAMPLES OF BRIDGE DESIGN AND REPLACING TEETH

Canines

The most challenging tooth to replace in the arch is the canine, particularly in the maxilla when the palatal surface is involved with lateral guidance. The adjacent teeth are normally the lateral incisor and the first premolar, neither of which is particularly large for retention of a bridge (Figure 19.26). In these situations alternative restorative options such as an implant or more extensive bridgework may need to be considered. The possibility of using double abutments (two or more retainers positioned on the same side as the edentulous space) for a cantilever design bridge has already been addressed and should be avoided wherever possible. Generally, double-abuting teeth increase the risk of failure as a pivot is established between the pontic and the closest retainer. Potential differential movement of the retainers on the abutment teeth creates the possibility of movement at the distal margin and theoretically increases the risk of partial decementation and consequently caries.



Figure 19.26 The canine pontic is cantilevered off the premolar tooth. The occlusion is being checked in lateral excursion to ensure the guidance is not on the canine pontic.

The other major consideration is the occlusion, particularly if the canine guides the mandible in lateral excursion. If the palatal surface of an upper canine pontic guides the mandible in lateral excursion, the stress placed upon either one (cantilever design) or two abutment (fixed-fixed or fixed-movable designs) teeth may exceed their retentive capacity. Even if small degrees of movement occur on the retainers, the risk of partial decementation and caries developing or complete decementation of the bridge increases. Generally, when planning bridges to replace canine teeth which have been involved in guidance, consideration should be given to changing the guidance to group function. This protects the retainers and the bridge but increases the complexity of the bridge, particularly when fitting the bridge in the mouth. Often adjustment of the opposing canine is necessary in order to produce an upper canine of adequate aesthetic length.

Premolar teeth, the shortened dental arch and removable prostheses

Distal cantilever designed bridges are normally considered to be a risk due to adverse occlusal forces and are consequently not commonly used. The one situation where a distal cantilever design is useful is when the opposing dentition is restored with a removable prosthesis. This is particularly relevant to the mandible when there is a shortened clinical arch. Research has shown that using bilateral free-end partial dentures to replace all the molar teeth is often unsuccessful. Provided at least one premolar tooth is present bilaterally, together with the canines and incisors, there is little functional benefit from replacing the molar teeth. Using the first premolars to support a distal cantilever designed conventional or resin retained



Figure 19.27 Distal cantilever bridge design. In the opposing arch is a complete removable prosthesis. The bridge design provides the patient with a shortened dental arch without the need for a lower removable prosthesis.

bridge will improve the function of the dentition without necessarily increasing risk of failure (Figure 19.27).

LABORATORY CONSTRUCTION OF BRIDGES

Most bridges are made from metal and ceramic. The metal provides the substructure and strength whilst the ceramic gives the shape and colour of the bridge. The metal needs to be of sufficient thickness to provide support and strength to the bridge; if it is too flexible the ceramic will undergo tension and crack. Therefore, a balance is needed between the thickness of the metal and sufficient space needed for the ceramic to hide the appearance of the metal. The ceramic is built up in a layering technique with opaque ceramic added first and then increasingly more translucent shades are added to provide the final appearance of the restoration (see Chapter 10 on metal–ceramic crowns).

With the advent of computer-aided design and computer-aided manufacture (CAD-CAM) technology and improvements in ceramics such as zirconia, all-ceramic bridges are now possible. The increased strength of the zirconia coping enables it to be used in bridge construction where the aesthetic demands are high and lifelike translucent restorations are required (Figure 19.28). To achieve an adequate strength between the pontic and the retainer, the connector for all ceramic bridges must be of sufficient size; as such, care needs to be taken to ensure adequate oral hygiene procedures. There is growing evidence that posterior all-ceramic bridges are more liable to mechanical failure whereas anterior bridges have a better prognosis.



Figure 19.28 All-ceramic six unit fixed–fixed Lava bridge. *(Courtesy of 3M ESPE)*

SUMMARY

Conventional bridges still remain an important part of the dentist's armamentarium for replacing missing teeth. Their use is mainly confined to situations where the abutment teeth are heavily restored. Preparation of teeth for full coverage retainers gives greater control over aesthetics in the anterior region and greater retention generally compared to resin retained bridges. Careful selection of bridge design is important and consideration needs to be given to the retainer, pontic and connector in addition to aspects of appearance (see Chapter 15). Tooth preparation (especially for fixed–fixed bridge designs) is demanding, not only from a mechanistic point of view but also from a biological and preservation of healthy tissues perspective.

FURTHER READING

Elder, A.R., Djemal, S., 2008. Electrosurgery: a technique for achieving aesthetic and retentive resin-bonded bridges. Dent. Update 35, 371–374, 376.

Lulic, M., Bragger, U., Lang, N.P., Zwahlen, M., Salvi, G.E., 2007. Ante's (1926) law revisited: a systematic review on survival rates and complications of fixed dental prostheses (FDPs) on severely reduced periodontal tissue support. Clin. Oral Implants Res. 18 (Suppl. 3), 63–72.

Pjetursson, B.E., Tan, K., Lang, N.P., Bragger, U., Egger, M., Zwahlen, M., 2004. A systematic review of the survival and complication rates of fixed partial dentures (FPDs) after an observation period of at least 5 years. Clin. Oral Implants Res. 15, 625–642.